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<Cultural Heritage Preservation Archive>

Test Plan Document

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1. Introduction

Software testing is a structured method for objectively assessing not only how the different components of a given software application should behave, but also if there are any potential issues in using those components. Testing creates confidence among the various parties involved in the development and use of an application that will be handled correctly throughout its life.

For the Cultural Heritage Preservation Archive System, software testing is a key component since it will be used to store, manage, and showcase collections of important cultural materials: e.g., artifacts, traditions, history, images, audio, and video. If a software application for managing important cultural information is improperly maintained, it can have a negative impact on many aspects of an end-user's ability to access this material, including the data's correctness, usability, and functional validity.

Testing is performed on the Cultural Heritage Preservation Archive System using a common methodology for testing both functional and non-functional aspects of an application, in order to ensure that the system performs consistently under different conditions and interactions with users. The effectiveness of this methodology is demonstrated by the ability of test results to provide measurable and observable results for evaluating the system's performance, quality, and reliability.

1.1 Purpose

The Test Plan has been developed with the intention of identifying testing scope, strategy and objectives for the new Cultural Heritage Preservation Archive System, as well as to improve the quality and reliability of the system through early identification and reporting of defects. Specifically, the objectives of this Test Plan are to:

- Validate that all functional requirements, such as browsing items of cultural interest, viewing details regarding an item, searching for items, and managing the content of the archive, function properly.
- Verify that the behavior of the Cultural Heritage Preservation Archive System is consistent across all interactions and usage scenarios.
- Verify that the data in the database is secure, accurate, and can be retrieved correctly.
- Verify that the Cultural Heritage Preservation Archive System satisfies basic non-functional requirements such as good usability, good performance, and high reliability.
- Minimize the potential risks associated with the Cultural Heritage Preservation Archive System by identifying defects prior to deployment.

Through this Test Plan, it is hoped that the testing performed on the system will help ensure that the final system functions as anticipated, provides a positive experience for users, and implements the long-term goal of preserving and making accessible the cultural heritage information in a digitized format in a manner that is reliable and dependable.

2. Features to be tested/not to be tested

2.1. Features to be tested

This section describes the features of the cultural heritage preservation system (CHPA) to be tested from the user's viewpoint and focusing on how different actors (viewers, contributors, and admins) interact with the system.

2.1.1. Functional Requirements

These features represent the core activities the user performs to preserve and access cultural heritage materials.

Table 1: Functional requirements

Feature Name	User Viewpoint	Risk Level
User registration and login	Ability of user to create account as viewer or Contributor and securely log into access specific Archive feature.	High (H)
Uploading Heritage item	A process of contributors to upload digital cultural Materials (includes oral traditions, artifacts and manuscripts)	High (H)
Search and view content	Ability of all users to browse, search and view heritage Items stored in the archive	High (H)
Metadata management	Ability for contributors to add and edit descriptive details For their uploaded items (dates, tags, locations)	Medium (M)
Content Moderation	The process of admins to review, approve or reject newly Uploaded heritage materials before going public	High (H)
User account management	This is the admin's ability to manage user roles, verify Contributors and maintain system access.	Medium (M)
Bookmark and favorites	Ability of user to save specific heritage items to them Personal collection for quick access later	Low (L)

Download and sharing	The functionality allowing users to download allowed Content or shared links to items via external platform	Medium (M)
Notifications	An alert sent to user regarding account update or status Of their content submission.	Low (L)

2.1.2. Non-Functional Requirements

These features describe the quality attributes of the CHPA system that ensure a reliable and secure experience of the user.

Table 2: Non-functional requirements

Category	Requirement Description	Risk Level
Security	ensuring that culturally sensitive data is protected and that only authorized users can perform restricted actions like uploading or deleting content.	High (H)
Availability	Ensuring the archive is accessible 24/7 so that users can View and contribute materials at any time.	High (H)
Usability	Ensuring that the interface is accessible for various user Types including those with various level of technical Expertise.	Medium (M)
Performance	This is the speed at which the heritage items are retrieved And displayed while browsing and searching.	Medium (M)
Data integrity and backup	Ensuring that one's cultural heritage data is uploaded, Correctly stored and can be restored if failure occurs	High (H)
Portability	The ability for the archive to function correctly across Different browsers and devices	Low (L)

2.1.3. Risk level justifications

This section is the justification to explain the risk rating assigned to the CHPA features.

High Risk (H)

- Registration, login and security: these are fundamental to the systems of integrity, meaning a failure here could lead to unauthorized access or the inability of users to utilize the archive at all.
- Upload and Moderation: Since the primary purpose of the CHPA is preservation, errors in the upload approval process and deletion can lead to the publication of inaccurate information and loss of irreplaceable cultural data.
- Data integrity and availability: the archive must be reliable and permanent. If data is corrupted or system is frequently down, then the preservation goal is compromised.

Medium Risk (M)

- Metadata management and search: while its essential for finding items, minor issues in those areas can make the system hard to use but not destroy the core data.
- Usability and performance: a slower system or slightly difficult interface affects the user experience but doesn't the actual preservation of heritage.

Low Risk (L)

- Bookmarks and notifications: these are convenience features, even if they fail users can still manually search for items and check their account status, resulting in minimal impact on primary goals of the archive.
- Portability: While broad device support is ideal, the archive remains functional for majority of users even if it has minor display issues on niche browsers.

2.2. Features not to be tested

This section describes the features of the Cultural Heritage Preservation Archive (CHPA) system that will not be tested in this release, based on how users interact with the system. The focus is on user-visible functions, not on internal or technical system behavior.

The listed features are excluded because they are not included in this release, considered low risk, or available but not formally tested or documented for this version of the software.

2.2.1 Administrative Actions Not Accessible to Regular Users

User View: Users cannot assign roles, manage other user accounts, or view system activity logs.

Reason Not Tested: These actions are not available to most users and are handled only by administrators. Since they are not part of normal user interaction, they are excluded from user-level testing in this release.

2.2.2 Advanced Content Review Decisions

User View: Users do not participate in detailed moderation processes such as content dispute resolution or appeal handling.

Reason Not Tested: Only basic approval and rejection actions are included in this version. Advanced moderation workflows are planned for future releases and are therefore not tested.

2.2.3 Content Rating and User Comments

User View: Users cannot rate cultural heritage items (for example, using star ratings) or leave public comments on archived content.

Reason Not Tested: This feature is not included in this release. To preserve cultural sensitivity and prevent inappropriate or disrespectful feedback, community interaction features are intentionally excluded from the initial version of the system.

2.2.4 User-to-User Messaging System

User View: Users cannot send private messages to other contributors or administrators within the system.

Reason Not Tested: This feature is considered low risk and has been used before through a stable, pre-existing messaging component. Since it is not a core archival function, it is excluded from active testing during this cycle.

2.2.5 Email and External Notifications

User View: Users may receive email notifications, but these notifications are not verified or tested in this release.

Reason Not Tested: Email notifications rely on external services and are considered at low risk. They are not formally tested or documented as part of this software version.

2.2.6 Heavy Usage and Peak-Time Scenarios

User View: Users will not experience or test system behavior during heavy traffic or large-scale usage.

Reason Not Tested: This system is intended for limited academic use in this release. High-volume usage scenarios are unlikely and are therefore excluded from testing

2.2.7 Different Devices and Browsers

User View: Users may not be able to access the system on all browsers, devices, or assistive technologies.

Reason Not Tested: Testing is limited to common environments. Full device compatibility and accessibility testing are planned for future versions.

2.2.8 Future and Optional User Features

User View: Some planned or optional features may not be visible or usable in this version.

Reason Not Tested: Only features officially included in this release are tested. Future enhancements will be tested when they are released.

3. Pass/Fail criteria

The pass/fail criteria define the standards used to determine whether the Cultural Heritage Preservation Archive system operates correctly and meets its stated requirements. Each test case will be evaluated based on functional correctness, data integrity, system response, and compliance with non-functional requirements.

A test case will be considered ***"Pass"*** when all the following conditions are satisfied:

- The system produces the expected output for the given input without errors or system crashes.
- User authentication and authorization function correctly (e.g., only authorized users can upload, edit, approve, or delete cultural heritage records).
- Cultural heritage records (text, images, audio, and video) are successfully uploaded, stored, retrieved, updated, and displayed as intended.
- Search and filtering operations return accurate and relevant results based on user queries.
- Input validation rules are enforced, including proper handling of empty fields, invalid data formats, and duplicate entries.
- The system responds within an acceptable time frame without noticeable delays when loading or accessing archive records.
- Security requirements such as protected access, session management, and restricted administrative actions are properly enforced.
- The user interface behaves consistently and displays clear messages for both successful operations and error conditions.

A test case will be considered ***"Fail"*** when any of the following occur:

- The actual output does not match the expected result defined in the test case.
- The system allows unauthorized access or performs restricted actions without proper permission.
- Cultural heritage data is not saved correctly, is lost, corrupted, or displayed incorrectly.
- Search results are inaccurate, incomplete, or unrelated to the user's query.
- Validation errors are not detected, or incorrect error messages are shown.

- System performance is significantly slower than expected, or the system becomes unresponsive.
- Security or data integrity requirements are violated.

All failed test cases will be recorded with details of the observed behavior, test data used, and the expected outcome. These failures will be reported to the development team for correction, and the affected test cases will be re-executed after fixes are applied to confirm that the system meets the required quality standards.

4. Approach/Strategy

4.1 Overall Testing Approach

Testing the Cultural Heritage Preservation Archive system focuses on correct data, trustworthy operation, and safe access to important digital assets. Because it holds sensitive cultural material, getting things right matters most alongside steady performance and proper permissions for who can view what.

A risk-based and incremental testing approach is adopted, aligned with the system development lifecycle. Testing activities begin early (shift-left testing) and continue throughout development to detect defects as early as possible.

The approach includes:

- Requirement based testing to validate functional and non-functional requirements
- Risk based testing focusing on critical modules such as artifact upload, metadata management, search and retrieval, and user authentication
- A combination of manual and automated testing to ensure both usability and functional correctness
- Role based testing to verify permissions for administrators, researchers, and general users
- Continuous testing during development and before each release

The following testing levels are applied:

- Unit Testing
- Integration Testing
- System Testing
- Regression Testing
- User Acceptance Testing (UAT)

4.2 Tools and Metrics

Tools:

The following tools are used to support development, testing, debugging, and quality assurance of the CHPA system:

- **Text Editors:** Visual Studio Code and Sublime Text for writing, debugging, and maintaining frontend and backend source code.

- **API & Backend Testing:** Postman for testing backend APIs, validating request and response data, and verifying authentication and authorization mechanisms.

- **Command-Line Tools:** PowerShell and Command Prompt (CMD) for running backend servers, managing databases, and executing scripts during testing.

- **AI-Assisted Tools:** Copilot and other AI tools for generating test cases, debugging support, code reviews, and improving overall test coverage.

- **Version Control:** Git integrated with GitHub repositories for source code management, collaboration, change tracking, and version control.

- **Browser Developer Tools:** Google Chrome Developer Tools (Console, Network, and Application tabs) for frontend debugging, monitoring API calls, inspecting storage, and analyzing performance issues.

Metrics:

Testing effectiveness and system quality are evaluated using the following metrics:

- Test case execution status (Pass, Fail, Blocked)
- Defect count categorized by severity (Critical, Major, Minor)
- Requirement coverage percentage
- API error rate and response status validation
- Page load time and network performance metrics

- Regression test pass rate
- Defect reoccurrence rate after fixes

These metrics help ensure the CHPA system remains reliable, secure, and maintainable.

System Development Environment (Technology Stack)

Core Frontend Stack

- React.js – Component-based UI development, routing, rendering.
- JavaScript (ES6+) – Logic, state management, API calls.
- HTML5 & CSS3 – Page structure and styling.
- Tailwind CSS – Utility-first styling, responsive layouts, professional UI components.
- Supporting Libraries
- React Router – Page navigation (Home, Explore, Profile, Login, Upload, Admin).
- Axios / Fetch API – Communication with backend APIs.
- React Icons – UI icons for upload, bookmark, admin, search, etc.

Core Backend Stack

- Node.js – JavaScript runtime environment.
- Express.js – Web framework for routing and middleware handling.
- Authentication & Authorization
- JSON Web Tokens (JWT) – Secure authentication.
- Role-Based Access Control (RBAC) – Enforce admin, contributor, and viewer permissions.
- bcrypt – Password hashing and security.
- Database & Data Management
- MySQL – Relational database for structured storage of users, roles, heritage items, bookmarks, approval status.
- MySQL Workbench / phpMyAdmin – Database design, query testing, table management.
- Media & File Handling
- Multer (Node.js middleware) – Upload images, audio, documents.
- Express Static Middleware – Serve uploaded media to frontend.

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Purpose

- Ensure robust testing coverage with modern QA tools and metrics.
- Build a responsive, secure, and user-friendly system using a modern frontend/backend stack.
- Provide persistent and structured data storage for cultural heritage items.
- Manage media files safely while maintaining accessibility and reliability.

4.3 Regression Testing Strategy

Regression testing is essential for the CHPA system due to frequent updates to archive content, metadata structures, and system features. The regression testing strategy includes:

- Maintaining a core regression test suite covering:
  - User authentication and authorization
  - Cultural artifact upload and validation
  - Metadata creation, editing, and storage,
  - Search, filtering, and retrieval functionality
  - Notification and approval workflows

- Automating regression tests for stable and high-risk functionalities where possible

- Executing regression testing:
  - After changes to metadata schemas
  - Following bug fixes or feature enhancements
  - Before each system release

Smoke testing is performed first to verify system stability, followed by full regression testing when required.

5. Test cases with specifications

These test cases focus on validating core system functionalities, user interactions, and role-based access control to ensure that the *Cultural Heritage Preservation Archive (CHPA)* system operates correctly according to its specified behavior.

Table 3: Test case specification – User registration

Name: User Registration				
Purpose: Allow new users to create an account a Viewer or Contributor				
Test Data = • Name (valid, invalid, empty) • Username (valid, already exits, empty) • Email (valid, invalid, empty) • Password (valid, weak, empty) • Confirm Password (match, mismatch)				
Input	**Expected result**	**Data**	**Actual output**	**Pass/fail**
All fields empty	Error: "All fields are required"	Name = " " Username = " " Email = " " Password = " "	Error message displayed	Pass
Invalid email format	Error: "Enter valid email"	Email="abc.com"	Error message displayed	Pass
Password and confirm password mismatch	Error message displayed	Password = "Pass123" Confirm = "Pass12"	Error message displayed	Pass
Username already exists	Error: "Username already taken"	Username = "existingUser"	Error message displayed	Pass
Weak password	Error: "Password must be strong"	Password = "Pass123"	Error message displayed	Fail

All valid inputs	Account created successfully	Name="Blain", Username=Blain01", Email="Blain@gmail.com", Password="Strong@123"	Account created successfully	Pass

Table 4: Test case specification - Login

Name = User Login				
Purpose = Authenticate user and create a secure session				
Test Data = • Username (valid, invalid, empty) • Password (valid, invalid, empty) • Password Storage Method (bcrypt-hashed) • Forgot Password Option (selected, not selected) • Security Question Answers (correct, incorrect) • New Password (valid, weak) • Confirm Password (match, mismatch) • Audio Playback (enabled, disabled)				
Input	**Expected result**	**Data**	**Actual output**	**Pass/fail**
Empty username/password	Error: "Can't leave empty"	Username=" " Password=" "	Error message displayed	Pass
Invalid credentials	Error: "Invalid username or password"	Username="Blain01" Password="wrong"	Error message displayed	Pass
Valid credentials (hashed password)	Login successful, redirect to dashboard	Username="Blain01" Password="Strong@123"	User redirected to dashboard	Pass
User selects Forgot Password	Redirected to authentication questions	User = "Blain01"	Authentication questions dispalyed	Pass
Forget Password - Incorrect answers	Error: "Authentication failed"	Answer = "Incorrect"	Authentication failure and message displayed	Pass
Forget Password - Correct answers	Allow password reset	Answer = "Correct"	Password reset page displayed	Pass
Reset password stored securely	Password saved as hashed value	NewPasword = "NewStrong@123"	Password stored using bcrypt hash	Pass

Audio playback enabled (Usability)	Login instructions read aloud	Audio = Enabled	Audio instructions played successfully	Pass

Table 5: Test case specification – Upload Cultural Heritage Items

Name: Upload Heritage Items				
Purpose: Allow Contributors/Admins to upload cultural heritage content				
Test Data = • Title (valid, empty) • Description (valid, empty) • File type (image/audio/video, invalid type) • Metadata (valid, invalid)				
Input	**Expected result**	**Data**	**Actual output**	**Pass/fail**
Viewer attempts upload	Access denied	Role = Viewer	Access denied and message displayed	Pass
Empty title	Error: "Title is required"	Title = " "	Validation error displayed	Pass
Empty metadata	Error: "Metadata is required"	Metadata = " "	Validation error displayed	Pass
Invalid file type	Error: "Unsupported file format"	File = "file.exe"	Upload blocked	Fail
Valid data by Contributor	Item saved as *Pending Approval*	Title="Ancient Manuscript", File="image.jpg"	Item stored with pending status	Pass
Large file size	Error: "File size exceeds limit"	File Size = "Set the limit"	Upload blocked	Pass

Table 6: Test case specification – Search Cultural Heritage Items

Name: Search Heritage Items
Purpose: Allow users to search archived content

Test Data =

- Keyword (valid, invalid, empty)
- Category filter (optional)

Input	Expected result	Data	Actual output	Pass/fail
Empty search query	Error: "Enter search keyword"	Keyword = " "	Validation error displayed	Pass
Valid keyword	Matching results displayed	Keyword = "Manuscript"	Relevant results shown	Pass
No matching data	Message: "No results found"	Keyword = "UnknownItem"	No results message displayed	Pass
Special characters only	Error: "Invalid search input"	Keyword = "@@@"	Error message displayed	Pass

Table 7: Test case specification – Metadata Management

Name: Metadata Management

Purpose: Allow Contributors/Admin to add and manage metadata for heritage items

Test Data =

- Author (valid, invalid, empty)
- Year (valid, invalid, empty)
- Region (valid, invalid, empty)
- Category (valid, invalid, empty)
- Metadata Field (editable, non-editable)
- User Role (Admin, Contributor, Viewer)

Input	Expected result	Data	Actual output	Pass/fail
Submit item with complete metadata	Metadata saved successfully	Author = "Unknown" Year = "1800" Region = "Ethiopia"	Metadata stored correctly	Pass
Submit item with missing requires metadata	Error: "Required metadata missing"	Region = " "	Validation error displayed	Pass

Edit metadata as Contributor	Metadata updated successfully	Role = Contributor Field = "Year"	Metadata updated	Fail
Viewer attempts metadata edit	Access denied	Role = Viewer	Access denied message displayed	Pass
Admin corrects metadata	Metadata updated successfully	Role = Admin Field = "Author"	Metadata updated by admin	Fail

Table 8: Test case specification – Approve/Reject Heritage

Name: Approve/Reject Item				
Purpose: Allow Admin to moderate submitted content				
Test Data = • Item status (pending) • Action (approve, reject)				
Input	**Expected result**	**Data**	**Actual output**	**Pass/fail**
Contributor attempts approval	Access denied	Role = Contributor	Access denied and message displayed	Pass
Admin approves item	Status changes to *Published*	Role = Admin Action = Approve	Item status updated to Published	Pass
Admin rejects item	Status changes to *Reject* + Notification sent	Role = Admin Action = Reject	Item rejected and notification sent	Pass
Admin approves already approved item	Error: "Action not allowed"	Status = Published	Error message displayed	Pass

Table 9: Test case specification – User Account Management

Name: User Account Management				
Purpose: Allow users to manage their account details and status				
Test Data = • Username (valid, invalid, inactive) • Email (valid, invalid, empty) • Old Password (correct, incorrect, empty) • New Password (valid, weak, empty) • Account Status (active, inactive) • Action (update profile, change password, deactivate account)				
Input	**Expected result**	**Data**	**Actual output**	**Pass/fail**
User updates profile with valid data	Profile updated successfully	Username = "Blain01" Email = "Blain@gmail.com"	Profile information updated	Fail
User updates profile with invalid email	Error: "Enter valid email"	Email = "abc@"	Validation error displayed	Fail
User changes password with wrong/old password	Error: "Incorrect current password"	Old password = "wrong123"	Error message displayed	Pass
User deactivates account	Account status set to inactive	Action = Deactivate	Account marked as inactive	Fail

Deactivated user attempts login	Login denied	Status = Inactive	Access denied message displayed	Pass

Table 10: Test case specification –Bookmark Item

Name: Bookmark Heritage Items				
Purpose: Allow users to save items for later access				
Test Data = • User Role (guest, logged-in user) • Item Status (published, unpublished) • Bookmark Action (add bookmark, duplicate bookmark) • Authentication Status (authenticated, unauthenticated)				
Input	**Expected result**	**Data**	**Actual output**	**Pass/fail**
Guest user bookmarks item	Login required	User = Guest	Redirected to login page	Pass
Logged-in user bookmarks item	Item added to bookmarks	User = Contributor	Item saved in bookmarks list	Pass

Table 11: Test case specification – Download/View Item

Name: Download/View Item				
Purpose: Allow users to access published cultural content				
Test Data = • User Role (viewer, contributor) • Item Status (published, pending, restricted) • Access Type (view, download) • File Type (image, audio, video) • Download Permission (allowed, denied)				
Input	**Expected result**	**Data**	**Actual output**	**Pass/fail**
Restricted item by Viewer	Access denied	Role = Viewer Item = Restricted	Access denied and message displayed	Pass
Published item	Content displayed/download	Item = Published	Content successfully displayed	Pass
Contributor views own pending item	Item displayed (read-only)	Role = Contributor	Item displayed as read- only	Pass

Table 12: Test case specification – Notification System

Name: Receive Notification				
Purpose: Inform users about approval, rejection, or updates				
Test Data = none				
Input	**Expected result**	**Data**	**Actual output**	**Pass/fail**
Item approved	Contributor receives notification	Action = Approve	Approval notification received	Pass
Item rejected	Contributor receives rejection notice	Action = Reject	Rejection notification received	Pass
Multiple items approved	Multiple notifications sent	Bulk approval	Approval notification received	Pass

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